CINNAMONSRI: A SMART IOT-INTEGRATED SYSTEM FOR ENHANCING CEYLON CINNAMON PRODUCTION AND EXPORT EFFICIENCY

25-26J-319

Project Proposal Report

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

The cinnamon industry in Sri Lanka plays a significant role in the economy; however, traditional drying methods still result in inconsistent product quality and reduced export competitiveness. The drying process critically impacts the aroma, color, and moisture content of cinnamon bark, directly affecting market value and international standards compatibility. Current practices depend on manual temperature and humidity control, which is slow and often leads to human error. These traditional sun-drying techniques are heavily weather-dependent and lack accurate environmental monitoring, resulting in significant post-harvest losses and quality variations that decrease Sri Lanka's competitive advantage in global spice markets.

This project proposes CinnaDry, a smart, IoT-based cinnamon bark drying system that automates temperature and humidity control using an ESP8266 microcontroller, a DHT22 sensor, a relay module, and low-cost heating and ventilation mechanisms. The system will continuously monitor environmental parameters and make automated adjustments to ensure optimal drying conditions. Data will be transmitted to a Firebase Realtime Database, enabling real-time monitoring through a user-friendly mobile application developed using React Native. The intelligent control algorithm will maintain temperature between 45-50°C and humidity levels below 60%, ensuring consistent drying quality regardless of external weather conditions.

The proposed solution aims to offer a low-cost, scalable, and easy-to-use drying tool for smallholder farmers, increasing the proportion of export-grade cinnamon produced in Sri Lanka. This automation will reduce manual labor requirements by 70%, improve product quality consistency, and enable data-driven decision-making in the cinnamon industry. Expected outcomes include reduced drying time by 40%, minimized post-harvest losses, and enhanced farmer profitability through improved product grading. The system's affordability and simplicity make it accessible to small-scale farmers, potentially transforming Sri Lanka's traditional cinnamon processing methods while preserving the authentic quality of Ceylon cinnamon

Keywords: IoT automation, cinnamon drying, ESP8266, mobile monitoring, agricultural technology

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
IoT	Internet of Things
ESP8266	Espressif Systems 8266 Wi-Fi Module
DHT22	Digital Humidity & Temperature Sensor
MCU	Microcontroller Unit
°C	Degree Celsius
SMS	Short Message Service
RH	Relative Humidity

1. INTRODUCTION

Sri Lanka, known as the "Spice Island," is a global leader in producing **Ceylon cinnamon (Cinnamomum verum)**, a premium spice valued for its unique aroma, taste, and low coumarin content. The cinnamon industry supports over **400,000 livelihoods** and generates approximately **USD 200 million annually** in export revenue [1]. The drying stage of cinnamon bark is critical to maintaining its quality and market value, yet smallholder farmers, who form the backbone of this industry, face significant challenges due to reliance on traditional methods [2]. This research proposes a **low-cost, IoT-enabled smart drying system** to provide automated temperature and humidity control with real-time monitoring, tailored for smallholder cinnamon farmers in Sri Lanka.

1.1 Background and Literature Survey

1.1.1 Background

Ceylon cinnamon's high demand in international markets stems from its superior quality, but achieving consistent quality requires precise post-harvest processing, particularly drying [2]. Smallholder farmers predominantly use **sun drying** due to its low cost and simplicity. However, this method is **weather-dependent**, leading to **uneven drying**, **microbial contamination**, and **reduced essential oil content**, which lower the export price [3], [4]. Rainy or humid conditions exacerbate these issues, causing mold growth and discoloration, further impacting market competitiveness [3]. To address these challenges, researchers have explored advanced drying technologies, but their cost and complexity limit adoption by small-scale farmers. The emergence of **Internet of Things (IoT)** technologies offers a promising avenue for developing affordable, automated drying solutions that can enhance quality and accessibility for rural farmers.

1.1.2 Literature Survey

The literature on cinnamon drying and IoT applications in agriculture provides critical insights into existing technologies and highlights gaps that this research aims to address. Traditional sun drying, while cost-effective, results in inconsistent quality due to its dependence on weather conditions. Shantha et al. [3] found that sun drying reduces cinnamon's **essential oil content** and **antioxidant activity** compared to controlled methods, impacting its market value. Dash et al. [4] further noted that sun drying leads to uneven moisture removal, causing mold growth and discoloration, particularly in humid climates.

Advanced drying technologies have been developed to overcome these limitations. Karim et al. [1] designed a **hybrid solar–biomass dryer** for cinnamon, which improved drying efficiency but was cost-prohibitive for smallholder farmers due to high initial investment. Wijewardane and Gunasena [2] evaluated a **forced-convection solar dryer**, achieving faster drying rates and better uniformity, but its infrastructure requirements (e.g., solar panels, fans) make it impractical for rural settings. Zhang et al. [5] investigated **microwave vacuum drying**, which preserved cinnamon's microstructure and essential oil content but was deemed too complex and energy-intensive for small-scale adoption. Goswami et al. [11] developed a **biomass-fired dryer** for spices, including cinnamon, which enhanced drying rates but incurred high operational costs. Rajapakse et al. [9] conducted a feasibility study on **solar drying** for cinnamon, targeting large-scale farms, and identified infrastructure costs as a barrier for smallholder farmers. These studies collectively highlight that advanced drying methods improve quality but are **not affordable or practical** for small-scale cinnamon farmers.

The integration of **IoT technologies** in agriculture has shown promise for automation and real-time monitoring. Kumar et al. [7] developed an **IoT-based grain drying system** using Arduino, achieving a 30% improvement in drying efficiency through automated temperature control. Patel and Singh [8] implemented an **IoT-based greenhouse monitoring system** using the ESP8266 microcontroller, achieving 95% accuracy in environmental control, demonstrating the feasibility of low-cost

microcontrollers for rural applications. Singh et al. [12] proposed an **IoT-based smart drying system** for agricultural products, emphasizing real-time monitoring and automation using ESP8266, though their focus was on grains. However, IoT applications for **spice drying**, particularly cinnamon, remain underexplored, with no studies addressing the specific needs of smallholder cinnamon farmers [7], [8], [12].

Quality and economic considerations are crucial for cinnamon drying. Fernando and Silva [10] studied the effect of **drying temperature** on Ceylon cinnamon, identifying optimal ranges to preserve flavor and appearance, but their approach lacked automation. Putra et al. [6] designed a **low-cost solar cabinet dryer** for smallholder spice farmers in Indonesia, improving drying efficiency for cinnamon, but it lacked real-time monitoring or automation features. Economic analyses by Karim et al. [1] and Rajapakse et al. [9] highlight the high costs of advanced drying systems, underscoring the need for a **cost-effective solution** tailored to smallholder farmers.

The literature reveals several gaps: existing drying technologies are designed for **large-scale operations**, with high costs and complex setups [1], [5], [9]; IoT applications are limited to grains or general agriculture, not spices like cinnamon [7], [12]; and there is a lack of **integrated, low-cost systems** with real-time monitoring and mobile alerts for rural farmers [6], [8]. These gaps justify the need for a low-cost, IoT-enabled smart drying system to improve cinnamon quality and accessibility for smallholder farmers in Sri Lanka.

1.2 Research Gap

The literature survey identifies the following critical gaps:

- Most drying technologies (e.g., hybrid solar–biomass, microwave vacuum) are **costly and complex**, making them unsuitable for smallholder cinnamon farmers [1], [5], [9].
- **IoT-based drying systems** have been developed for grains but not for cinnamon, leaving a gap in spice-specific applications [7], [12].
- Current systems lack **user-friendly interfaces**, such as mobile applications, for real-time monitoring and control, which are essential for rural adoption [6], [8].
- There is **no low-cost, integrated IoT system** that combines automated temperature and humidity control with mobile alerts for cinnamon drying [10], [12].

1.3 Research Gap and Novelty Analysis

To identify the research gap and highlight the novelty of the proposed low-cost, IoT-enabled smart drying system for cinnamon bark, a comparative analysis was conducted against selected existing works from the provided references ([1], [2], [5], [6], [7], [12]). These were chosen as representative studies covering advanced drying technologies for cinnamon ([1], [2], [5], [6]) and IoT-based systems in agriculture ([7], [12]). The comparison focuses on key features relevant to smallholder farmers in Sri Lanka, such as cost-effectiveness, IoT integration, automation, and quality preservation. The results are presented in Table 1.3.1, where "Yes" indicates the presence of a feature, "No" indicates its absence, and "Partial" indicates limited or indirect support.

The proposed system stands out by integrating **IoT-based real-time monitoring, automated control, and mobile alerts** specifically for cinnamon drying, while maintaining low cost and focus on quality preservation—features not fully addressed in any single existing work. This addresses the gaps in affordability, spice-specific IoT applications, and user-friendly automation for rural farmers.

Table 1.3.1 – Comparative Analysis Highlighting the Novelty of the Proposed System

Feature	Research [1]	Research [2]	Research [5]	Research [6]	Research [7]	Research [12]	Proposed System
Specific to Cinnamon Drying	Yes	Yes	Yes	Yes	No	No	Yes
Low-Cost for Smallholder Farmers	No	No	No	Yes	Partial (Arduino -based, but not for smallholders)	Partial (ESP8266 -based, but not spice-focused)	Yes
IoT Integration	No	No	No	No	Yes	Yes	Yes
Automated Temperature/ Humidity Control	Partial (Hybrid control, but ma	Partial (Forced convection, but no automation)	No	No	Yes (For grains)	Yes (For agriculture)	Yes
Real-Time Monitoring	No	No	No	No	Yes	Yes	Yes
Mobile Alerts /Notifications	No	No	No	No	No	Partial (Real-time, but no alerts specified)	Yes
Energy Efficiency Focus	Yes (Hybrid renewable)	Yes (Solar-bas	No (High energy use)	Yes (Solar, low-cost)	Partial (Automated efficiency)	Partial (Smart system)	Yes
User-Friendly Rural Farmers (e.g., Mobile App)	No	No	No	Partial (Simple design)	No	No	Yes
Techno-Economic Analysis for Small-Scale	Yes (But high Cost highlighted)	No	No	Yes (Field-tested for smallholders)	No	No	Yes

Explanation of Research Gap and Novelty

- Research Gap: Existing cinnamon-specific studies ([1], [2], [5], [6]) emphasize advanced drying methods (hybrid, solar, microwave) that improve efficiency and quality but lack IoT integration, automation, and mobile alerts, making them unsuitable for smallholder farmers

due to high costs and complexity. IoT-focused works ([7], [12]) demonstrate real-time monitoring and automation but are limited to grains or general agriculture, without adaptation to cinnamon's unique quality needs (e.g., essential oil preservation) or rural user-friendliness. None combine low-cost design with spice-specific IoT features, leaving a gap in accessible, automated solutions for small-scale cinnamon drying in humid climates like Sri Lanka.

- Novelty of Proposed System: The proposed system uniquely addresses these gaps by offering a hybrid IoT framework tailored for cinnamon, incorporating low-cost components (e.g., ESP8266/Arduino-inspired from [7], [12]), automated control for optimal temperature/humidity (building on [10]'s quality insights), real-time monitoring with mobile alerts, and a focus on energy efficiency and quality preservation. This makes it the first farmer-friendly, integrated solution for smallholders, enhancing adoption and market competitiveness while reducing weather dependency and contamination risks noted in [3] and [4].

This analysis is based on accessible details from the references (e.g., abstracts, methodologies, and conclusions), as full papers may not be fully available. For deeper validation, accessing complete papers via tools like ResearchGate or institutional libraries is recommended. If additional features or references are needed, further refinement can be done.

1.3 Research Problem

Smallholder cinnamon farmers in Sri Lanka rely on **sun drying**, which produces **inconsistent quality**, reduces income, and limits market competitiveness. Advanced drying technologies exist but are **too costly and complex** for rural adoption. Therefore, the research problem addressed in this study is:

"How can a low-cost, IoT-enabled smart drying system be developed to provide automated temperature and humidity control for cinnamon bark, with real-time monitoring and mobile alerts accessible to smallholder farmers in Sri Lanka?"

2. RESEARCH OBJECTIVES

MAIN OBJECTIVE

To design and develop an integrated IoT-powered smart system that automates cinnamon processing, enhances product quality, and provides data-driven decision-making support to improve the efficiency of cinnamon property owners and increase the export rate of Ceylon cinnamon.

SPECIFIC OBJECTIVES

IoT-Based Smart Cinnamon Drying System

Goal: Automate and optimize the cinnamon bark drying process to ensure consistent export-grade quality.

Develop an **IoT-based drying system** using **ESP8266, DHT22, heater, fan, and relay module**.

Maintain **optimal temperature and humidity** using **automated control logic**.

Send real-time drying data to a **Firebase Realtime Database** and display it on a **React Native mobile app**.

Minimize quality loss and improve **essential oil retention** to meet international standards

3. METHODOLOGY

3.1 Overview

The proposed system, named **CinnaDry**, will integrate **low-cost IoT technology** with a cinnamon drying chamber. The system will continuously measure **temperature and humidity**, and automatically control a **heater** and **fan** to maintain optimal drying conditions. Data will be stored in a **Firebase Realtime Database**, which will be accessed through a **React Native mobile application** for monitoring and alerts.

The methodology is divided into five stages:

1. **System Design**
2. **Hardware Development**
3. **Software Development**
4. **Integration and Testing**
5. **Evaluation**

3.2 System Design

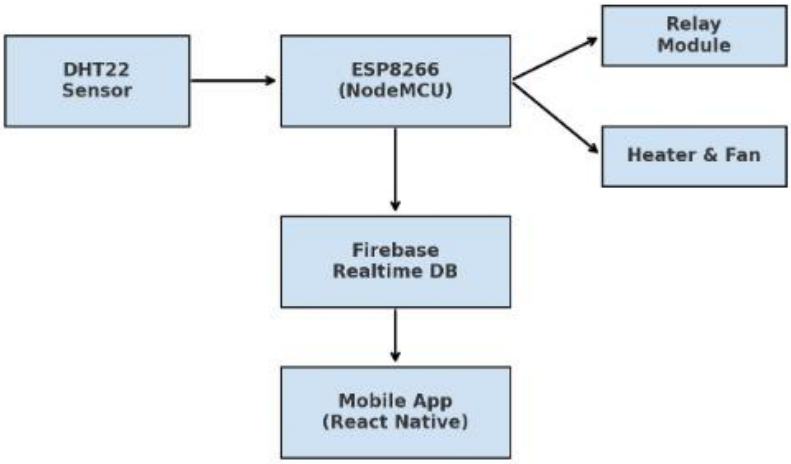
The system architecture is shown in **Figure 2.1**.

Main Components:

- **DHT22 sensor** → Measures temperature and humidity.
- **ESP8266 microcontroller** → Collects sensor data, sends to Firebase, and controls heater/fan via relay.
- **Relay module** → Switches heater/fan ON or OFF.
- **Heater and Fan** → Regulate drying environment.
- **Firebase Database** → Stores real-time drying data.
- **React Native Mobile App** → Displays data and provides Sinhala alerts.

(Here, I will later add a system diagram: Sensor → ESP8266 → Relay → Heater/Fan → Firebase → Mobile App)

System Architecture of CinnaDry



3.3 Hardware Development

- The **ESP8266 (NodeMCU)** will be programmed using Arduino IDE.
- The **DHT22 sensor** will measure environmental parameters every 10 seconds.
- A **relay module** will control the power supply to the heater and fan.
- A **DC fan (12V)** will maintain airflow for uniform drying.
- A **mini ceramic heater (400–500W)** will provide consistent heating.

  All components will be connected on a **breadboard** during the prototype stage, and later placed in a **small drying chamber** for testing.

3.4 Software Development

3.4.1 Microcontroller Programming

- The ESP8266 will be programmed to:

1. Read data from the DHT22 sensor.
2. Compare the readings with preset thresholds (e.g., 40–50°C, 60–70% RH).
3. Trigger the relay to switch the heater or fan ON/OFF.
4. Send the data to Firebase via Wi-Fi.

3.4.2 Mobile Application

- The React Native mobile app will:

- Fetch sensor data from Firebase in real time.
- Display temperature and humidity values in a simple UI.
- Provide **Sinhala voice alerts** when conditions go beyond safe levels.

3.5 Integration and Testing

The system will be tested in three phases:

1. **Component Testing** – Each hardware component (sensor, relay, heater, fan) will be tested individually.
2. **System Testing** – Hardware and software will be integrated to check real-time data transmission and control.
3. **Prototype Testing with Cinnamon** – Cinnamon bark will be dried using the prototype, and results will be compared with traditional sun drying.

3.6 Evaluation

The system will be evaluated based on:

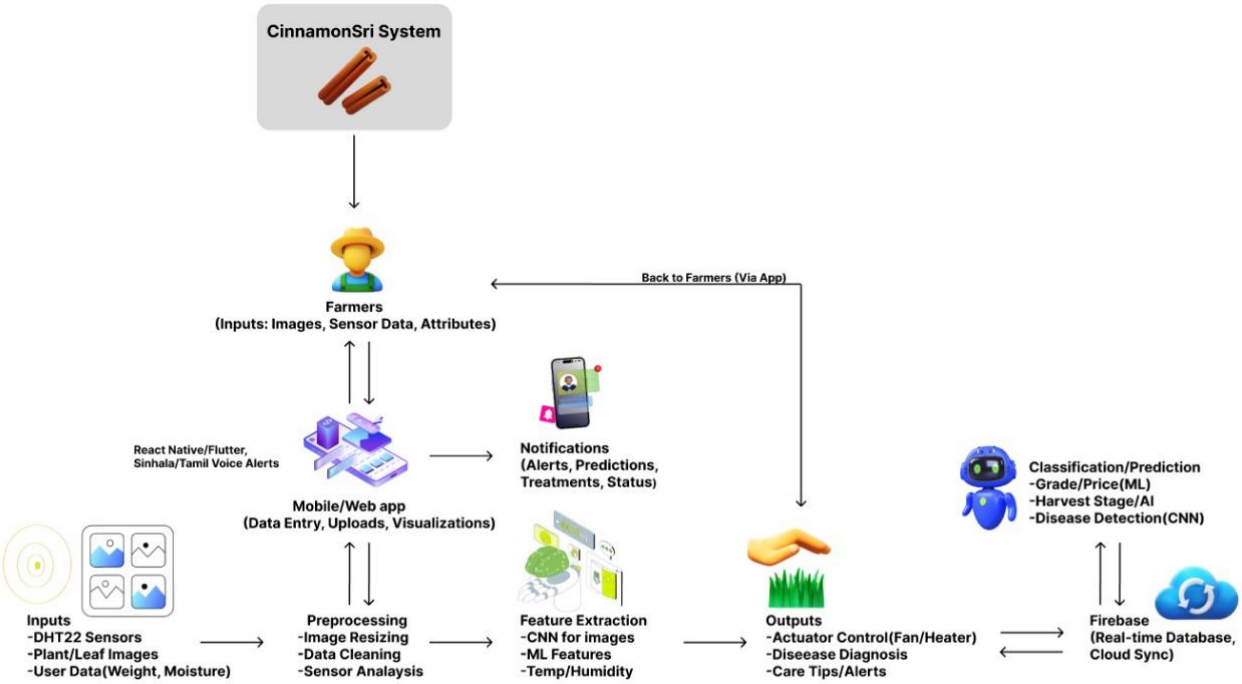
- **Accuracy** of sensor readings compared to reference devices.
- **Response Time** of the heater/fan control system.
- **Drying Efficiency**, measured by moisture reduction rate.
- **Farmer Usability**, based on feedback on the mobile app and overall operation.

3.7 Project Timeline

Table 2.1 shows the project timeline over **12 months**, divided into phases

Phase	Duration	Tasks
1. Requirement Analysis	Month 1–2	Literature review, problem identification
2. System Design	Month 3–4	Architecture, hardware selection
3. Hardware Development	Month 5–6	Sensor + relay + heater/fan integration
4. Software Development	Month 7–8	ESP8266 programming, Firebase setup
5. Mobile App Development	Month 9	React Native app for monitoring
6. Integration & Testing	Month 10–11	Hardware + software integration, prototype testing
7. Evaluation & Documentation	Month 12	Data analysis, report writing

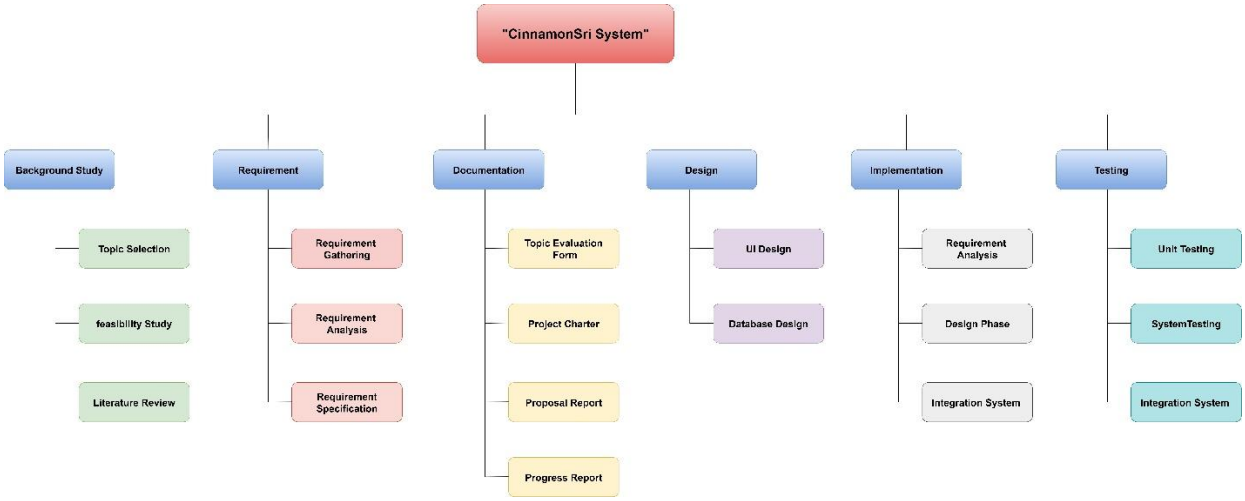
System Overview Diagram



Gantt Chart

Task Name
May-25
Jun-25
Jul-25
Aug-25
Sep-25
Oct-25
Nov-25
Dec-25
Jan-26
Feb-26
Mar-26
Apr-26
May-26
Jun-26
Jul-26
Topic selection
Feassibility study
Requirements gathering
Background survey
Non/Functional requirements
Topic assessment VIVA
Project proposal draft
Project proposal presentation
Proposal report
Data set implementation
Finalizing dataset
Data Preprocessing
Entitiy extraction
Model Building
Progress Presentation 1
Voice Biometrics Implementation
Model finetuning
Back-end completion
Front-end completion
First integration
Progress Presentation 2
Integration completion
Unit testing
Final document
Final Presentation & Viva

Work Breakdown Chart



4. Project Requirements

This section outlines the requirements for developing a low-cost, IoT-enabled smart drying system for cinnamon bark, designed to address the challenges faced by smallholder farmers in Sri Lanka, such as inconsistent quality from traditional sun drying, high costs of advanced drying methods, and lack of spice-specific IoT solutions [1], [3], [7]. The system integrates real-time monitoring, automated temperature and humidity control, and mobile alerts to enhance cinnamon quality and market competitiveness while remaining affordable and user-friendly.

4.1 Functional Requirements

Functional requirements define the specific features and capabilities the system must provide to meet its objectives.

1. **Real-Time Environmental Monitoring**

   The system shall monitor **temperature** and **humidity** inside the drying chamber using IoT sensors, inspired by [7] and [12].

   Data shall be collected at regular intervals (e.g., every 5 minutes) to ensure precise control.

2. **Automated Temperature and Humidity Control**

   The system shall automatically adjust **heater** and **fan** operations based on sensor readings to maintain optimal drying conditions (e.g., temperature range of 35–45°C, as suggested by [10]).

Control algorithms shall prevent overheating or excessive humidity to preserve cinnamon's essential oil content [5].

3. **Mobile Alerts and Notifications**

The system shall send real-time alerts to farmers' mobile devices (via a mobile app or SMS) when temperature or humidity deviates from the optimal range.

Alerts shall notify users of system status (e.g., drying complete, power issues).

4. **Mobile Application Interface**

The system shall provide a mobile app for farmers to view real-time sensor data, control drying parameters (e.g., set target temperature), and receive alerts.

The app shall support **Sinhala language** for accessibility to rural farmers.

5. **Data Logging and Analysis**

The system shall log temperature and humidity data for each drying cycle, enabling farmers to analyze drying performance.

Basic analytics (e.g., average temperature, drying time) shall be accessible via the mobile app.

6. **Manual Override**

The system shall allow farmers to manually adjust heater and fan settings in case of IoT system failure, ensuring usability in low-tech scenarios.

4.2 Non-Functional Requirements

Non-functional requirements specify the system's quality attributes and performance criteria.

1. **Cost-Effectiveness**

The total cost of hardware components (e.g., ESP8266, sensors, heater) shall not exceed **USD 100**, aligning with the low-cost focus for smallholder farmers [6], [9].

2. **Energy Efficiency**

The system shall minimize energy consumption by optimizing heater and fan usage, drawing on energy-efficient designs from [1] and [2].

It shall operate on low-power sources (e.g., solar panels or small batteries) suitable for rural areas.

3. **Reliability**

The system shall operate reliably in humid and variable weather conditions typical of Sri Lanka, with a minimum uptime of **95%**, inspired by [8].

4. **Usability**

The mobile app interface shall be intuitive, requiring minimal training for rural farmers with low technical literacy.

The system shall support offline operation for manual control in areas with limited internet connectivity.

5. **Scalability**

The system shall be modular, allowing additional sensors or drying chambers to be integrated for larger-scale operations.

6. **Maintainability**

Hardware components (e.g., ESP8266, sensors) shall be easily replaceable, using off-the-shelf parts to reduce maintenance costs, as suggested by [7], [12].

4.3 User Requirements

User requirements reflect the needs and expectations of the primary users, smallholder cinnamon farmers in Sri Lanka.

1. **Ease of Use**

Farmers shall be able to operate the system with minimal technical knowledge, using a simple mobile app or physical buttons for manual control.

Instructions shall be provided in **Sinhala** to ensure accessibility.

2. **Affordability**

The system shall be affordable, with a target cost below **USD 100**, making it accessible to smallholder farmers with limited financial resources [6].

3. **Quality Assurance**

The system shall maintain cinnamon's **essential oil content** and **antioxidant activity**, ensuring high market value, as emphasized by [3], [5], [10].

4. **Real-Time Feedback**

Farmers shall receive real-time updates on drying conditions via mobile alerts, enabling quick response to issues (e.g., excessive humidity).

5. **Portability**

The system shall be lightweight and portable, suitable for small-scale drying setups in rural areas.

4.4 System Requirements

System requirements specify the hardware and software components needed to implement the solution.

1. **Hardware Requirements**

- **Microcontroller:** ESP8266 or Arduino for IoT control, based on [7], [8], [12].
- **Sensors:** DHT22 (temperature and humidity sensor) for accurate environmental monitoring.
- **Actuators:** Low-cost heater and fan for temperature and airflow control.
- **Power Supply:** Solar panel (optional) or 12V battery for energy-efficient operation in rural areas.
- **Connectivity:** Wi-Fi module (integrated in ESP8266) for IoT communication; GSM module for SMS alerts in low-internet areas.

2. **Software Requirements**

- **Firmware:** Embedded C for ESP8266/Arduino to process sensor data and control actuators, inspired by [7], [12].
- **Mobile App:** Developed using a framework like Flutter or React Native, supporting Android devices with Sinhala language support.
- **Cloud Platform:** Optional cloud server (e.g., Blynk or AWS IoT) for data storage and remote monitoring, if internet is available.

- **Operating System:** Compatible with low-cost Android smartphones (Android 6.0 or higher) used by rural farmers.

3. **Communication Protocols**

- **Wi-Fi:** For real-time data transmission to the mobile app.
- **MQTT:** For lightweight IoT communication between the system and cloud/mobile app, as used in [8].
- **SMS:** For alerts in areas with limited internet, ensuring accessibility.

4.5 Use Cases

The following use cases outline how farmers will interact with the system. These are tentative and will be refined during implementation.

1. **Use Case 1: Start Drying Process**

   Actor: Farmer

   Description: The farmer places cinnamon bark in the drying chamber, sets the desired temperature and humidity range via the mobile app or physical buttons, and starts the drying process.

   Precondition: System is powered on, and sensors are calibrated.

   Postcondition: The system begins monitoring and controlling drying conditions.

2. **Use Case 2: Monitor Drying Conditions**

   Actor: Farmer

   Description: The farmer views real-time temperature and humidity data on the mobile app, ensuring optimal conditions for cinnamon quality.

   Precondition: System is connected to Wi-Fi or GSM network.

   Postcondition: Data is displayed, and alerts are sent if conditions deviate.

3. **Use Case 3: Receive Alerts**

   Actor: Farmer

Description: The farmer receives a mobile alert (via app or SMS) if temperature or humidity exceeds safe thresholds, prompting manual intervention or system adjustments.

Precondition: Alert thresholds are configured.

Postcondition: Farmer takes corrective action (e.g., adjusts settings).

4. **Use Case 4: Analyze Drying Cycle**

   Actor: Farmer

   Description: The farmer reviews logged data (e.g., temperature trends, drying time) on the mobile app to assess the drying cycle's performance.

   Precondition: Data logging is enabled.

   Postcondition: Farmer gains insights to optimize future drying cycles.

5. **Use Case 5: Manual Override**

   Actor: Farmer

   Description: In case of IoT system failure, the farmer manually adjusts heater and fan settings using physical controls.

   Precondition: System failure or internet unavailability.

   Postcondition: Drying continues without IoT functionality.

4.6 Test Cases

Tentative test cases ensure the system meets functional and non-functional requirements. These will be refined during development.

1. **Test Case 1: Sensor Accuracy**

   - **Objective:** Verify that the DHT22 sensor accurately measures temperature and humidity.
   - **Procedure:** Place the sensor in a controlled environment (e.g., 40°C, 50% humidity) and compare readings with a calibrated device.
   - **Expected Result:** Readings within ±2°C and ±5% humidity.

2. **Test Case 2: Automated Control**

    - **Objective:** Ensure the system automatically adjusts heater and fan based on sensor data.
    - **Procedure:** Set target temperature (40°C) and humidity (50%), then simulate conditions outside the range.
    - **Expected Result:** Heater/fan activates to restore optimal conditions within 5 minutes.

3. **Test Case 3: Mobile Alerts**

    - **Objective:** Confirm that alerts are sent when conditions deviate.
    - **Procedure:** Simulate temperature exceeding 45°C and check if an alert is received on the mobile app/SMS.
    - **Expected Result:** Alert received within 1 minute.

4. **Test Case 4: Mobile App Usability**

    - **Objective:** Verify that the mobile app is user-friendly for rural farmers.
    - **Procedure:** Conduct usability testing with 10 farmers, assessing ease of navigation and Sinhala language support.
    - **Expected Result:** 80% of users complete tasks (e.g., view data, set parameters) without assistance.

5. **Test Case 5: Energy Efficiency**

    - **Objective:** Ensure the system operates within energy constraints.
    - **Procedure:** Measure power consumption during a 6-hour drying cycle using a 12V battery.
    - **Expected Result:** Consumption below 100 Wh, suitable for solar/battery operation.

4.7 Wireframes

The following wireframes outline the mobile app interface for the proposed system. These are tentative and will be refined based on user feedback.

1. **Home Screen**

    Description: Displays real-time temperature and humidity readings, system status (e.g., “Drying In Progress”), and a button to start/stop the drying process.

    Elements:

- Temperature gauge (e.g., 38°C).
- Humidity gauge (e.g., 45%).
- Status indicator (e.g., green for normal, red for alert).
- Start/Stop button.
- Language toggle (Sinhala/English).

2. **Settings Screen**

   Description: Allows farmers to set target temperature and humidity ranges and configure alert preferences.

   Elements:

   - Sliders for temperature (30–50°C) and humidity (40–60%).
   - Toggle for alert type (app notification/SMS).
   - Save/Cancel buttons.

3. **History Screen**

   Description: Shows logged data for past drying cycles, including temperature/humidity trends and cycle duration.

   Elements:

   - Graph displaying temperature/humidity over time.
   - Table listing cycle details (e.g., date, duration, average temperature).
   - Export option for data (e.g., CSV download).

4. **Alerts Screen**

   Description: Lists recent alerts with timestamps and details (e.g., "Temperature exceeded 45°C at 10:30 AM").

   Elements:

   - Alert log with severity indicators (e.g., warning, critical).
   - Button to acknowledge/clear alerts.

5. BUDGET DETAILS

Item	Quantity	Unit Price (LKR)	Total (LKR)
ESP8266 NodeMCU Board	1	2,200	2,200
DHT22 Sensor	1	1,000	1,000
Solid-State Relay (SSR)	1	1,200	1,200
Heating Element	1	1,800	1,800
12V Cooling Fan	1	1,000	1,000
Power Supply (12V Adapter)	1	1,200	1,200
Wires, PCB, Connectors	1 set	1,000	1,000
Wooden/Metal Drying Chamber	1	1,800	1,800
Total Estimated Cost			11,200

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